

Conclusion and Future Work

Conclusion

This thesis presents a design and implementation of a web-based examination management application that is fast, secure, and easy to use. The system is built with modern technologies and clean architectural patterns to provide a reliable, fast, and scalable solution for managing examinations. The system is solving the problems, such as:

- **Automated grading and scoring:** The system grading process is automatic, with the ability to download the results as a PDF file.
- **Data integrity:** The integrity of the system is ensured by database constraints, the application layer, and HTTP middleware.
- **Security:** The system is designed with OWASP Top 10 vulnerabilities in mind. Each vulnerability is addressed with a specific solution.
- **Efficient resource utilization:** Go as the main language of the system delivers small binary files, is memory efficient, and has a fast startup. Combined with a type-safe template engine (Templ), a server-side rendering (SSR), and a lightweight interactivity framework (HTMX).
- **Scalability and Maintainability:** The system is designed with the Hexagonal Architecture design pattern, which is a scalable and maintainable solution.

Future Work

1. Proctoring and Anti-Cheating Mechanisms
2. Advanced Question Types
3. Mobile-Responsive Application